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Roman Numerals

Introduction

The system of Roman numerals dates back to the Roman civilization that existed long before the Hindu-Arabic numerals came into being. Though Roman numerals are now not used in mathematical calculations, yet we may sometimes notice them in watches, clocks and calendars.

There are seven basic symbols to write any Roman numeral.

These symbols with their corresponding Hindu-Arabic numerals are given below:



Roman Numeral	I	V	X	L	C	D	M
Hindu-Arabic Numeral	1	5	10	50	100	500	1000

In the Roman system, there is no symbol for zero. This is also not a place value system.

Using the above seven symbols, we may form Roman numerals for all numbers by adopting certain rules. However, here we shall discuss the rules for usage of the first three symbols - I, V and X - and form Roman numerals up to 39.

Rule 1: Repetition of a Roman numeral means addition.

Examples:

$$II = 1 + 1 = 2, III = 1 + 1 + 1 = 3,$$

$$XX = 10 + 10 = 20, XXX = 10 + 10 + 10 = 30.$$

Caution: (1) Only I and X can be repeated.

The numeral V is never repeated.

(2) No numeral can be repeated more than three times.



Rule 2: A smaller numeral written to the right of a larger numeral is always added to the larger numeral.

Examples:

$$VI = 5 + 1 = 6, VII = 5 + 1 + 1 = 7, VIII = 5 + 1 + 1 + 1 = 8,$$

$$XI = 10 + 1 = 11, XII = 10 + 1 + 1 = 12, XIII = 10 + 1 + 1 + 1 = 13.$$

Rule 3: A smaller numeral written to the left of a larger numeral is always subtracted from the larger numeral.

Examples:

$$IV = 5 - 1 = 4, IX = 10 - 1 = 9.$$

Caution: (1) I can be subtracted from V and X.

(2) The numeral V is never subtracted.

Rule 4: For numbers beyond 10, we first write the number in groups of 10s and 1s and then form the Roman numeral corresponding to the given number.

Examples:

$$11 = 10 + 1 = \text{XI},$$

$$12 = 10 + 2 = \text{XII},$$

$$13 = 10 + 3 = \text{XIII},$$

$$14 = 10 + 4 = \text{XIV},$$

$$15 = 10 + 5 = \text{XV},$$

$$16 = 10 + 6 = \text{XVI},$$

$$20 = 10 + 10 = \text{XX},$$

$$24 = 10 + 10 + 4 = \text{XXIV},$$

$$30 = 10 + 10 + 10 = \text{XXX},$$

$$39 = 10 + 10 + 10 + 9 = \text{XXXIX}.$$

Using the above rules, we may write the Roman numerals for Hindu-Arabic numerals from 1 to 39 as shown below:

Hindu-Arabic Numerals	Roman Numerals	Hindu-Arabic Numerals	Roman Numerals	Hindu-Arabic Numerals	Roman Numerals	Hindu-Arabic Numerals	Roman Numerals
1	I	11	XI	21	XXI	31	XXXI
2	II	12	XII	22	XXII	32	XXXII
3	III	13	XIII	23	XXIII	33	XXXIII
4	IV	14	XIV	24	XXIV	34	XXXIV
5	V	15	XV	25	XXV	35	XXXV
6	VI	16	XVI	26	XXVI	36	XXXVI
7	VII	17	XVII	27	XXVII	37	XXXVII
8	VIII	18	XVIII	28	XXVIII	38	XXXVIII
9	IX	19	XIX	29	XXIX	39	XXXIX
10	X	20	XX	30	XXX		

Since X cannot be repeated more than three times, we cannot write 40 as XXXX.

So, we represent 40 in Roman numerals as under:

$$40 = 50 - 10 = \text{XL}.$$



Exercise 5

1. Write the Roman numerals for the following.

- | | | | | |
|--------|--------|--------|--------|--------|
| (a) 6 | (b) 8 | (c) 12 | (d) 14 | (e) 26 |
| (f) 28 | (g) 33 | (h) 34 | (i) 38 | (j) 39 |

2. Write the Hindu-Arabic numerals for the following.

- | | | | | |
|------------|----------|----------|-----------|-----------|
| (a) IX | (b) XVI | (c) XXVI | (d) XIX | (e) XXXVI |
| (f) XXVIII | (g) XIII | (h) XXIX | (i) XVIII | (j) XX |

3. Fill in the placeholders with correct symbol $>$, $<$ or $=$.

- | | |
|----------------------------------|----------------------------------|
| (a) XXVI <input type="text"/> 24 | (b) XI <input type="text"/> 9 |
| (c) XIX <input type="text"/> 19 | (d) XXXV <input type="text"/> 34 |
| (e) X <input type="text"/> 11 | (f) XXIX <input type="text"/> 29 |

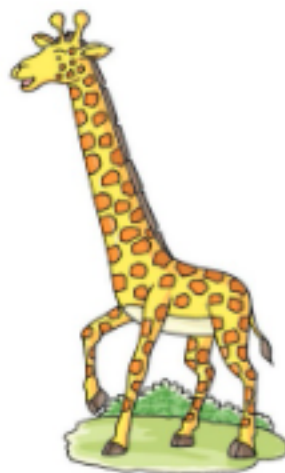


4. Which of the following are meaningless?

- | | | | |
|---------|----------|----------|----------|
| (a) XIX | (b) IXVI | (c) VV | (d) XX |
| (e) VX | (f) XIV | (g) XIIV | (h) XIII |

5. Solve and write the answers in Roman numerals.

- | | | |
|-------------|----------------|-----------------|
| (a) XI + XI | (b) VI + IV | (c) VII + XVIII |
| (d) XI - V | (e) XVI - VIII | (f) XXX - XXI |





Activity Time



Take a box of matchsticks and try to form Roman numerals for all Hindu-Arabic numerals from 1 to 40. Some numerals have been shown below to guide you in the formation process.



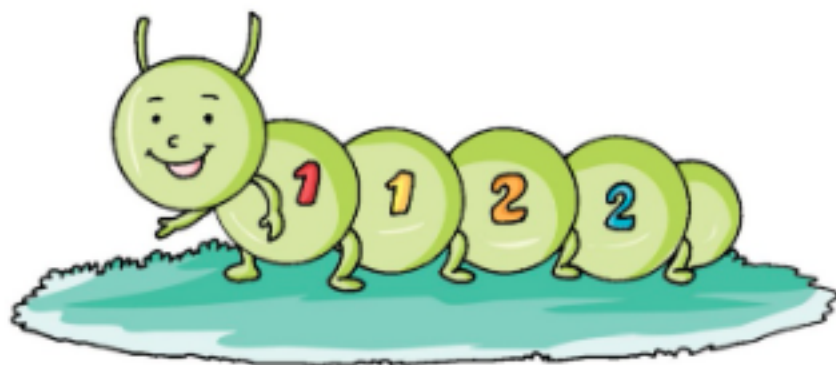
Count the number of matchsticks needed to form each numeral and then prepare a chart mentioning the numeral and number of matchsticks required as shown below:

Hindu-Arabic Numeral	Roman Numeral	Number of Matchsticks used
1	I	
2	II	
3	III	
4	IV	
5	V	

and so on.

Which number required maximum number of matchsticks?

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**QUESTION BAG 1****(Objective Type Questions)****Tick (✓) the correct answer.**

1. The number which cannot be represented by Roman numerals is
(a) 0 ☐ (b) 11 ☐ (c) 99 ☐ (d) None of these ☐
2. The Roman numeral for 8 is
(a) IIX ☐ (b) IIIIIII ☐ (c) VIII ☐ (d) None of these ☐
3. In Roman numerals, X can be used at the most
(a) 1 time ☐ (b) 2 times ☐ (c) 3 times ☐ (d) 4 times ☐
4. The Roman numeral that represents 100 is
(a) L ☐ (b) C ☐ (c) D ☐ (d) M ☐
5. Which one of the following is the correct statement?
(a) $XXV < 22$ ☐ (b) $XXL = 30$ ☐ (c) $XXIX < 30$ ☐ (d) $LL = 100$ ☐
6. Which of the following is a valid Roman numeral?
(a) IXIV ☐ (b) XIXX ☐ (c) XVX ☐ (d) XIX ☐
7. The 'V' in Roman numerals can be represented only
(a) once ☐ (b) twice ☐ (c) thrice ☐ (d) four times ☐
8. $XVI + VII = ?$
(a) XXIII ☐ (b) XXIV ☐ (c) XXVI ☐ (d) XXVII ☐
9. $XXX - XV = ?$
(a) XX ☐ (b) XXV ☐ (c) XV ☐ (d) X ☐
10. The correct Roman numeral for 17 is
(a) VVVII ☐ (b) XVII ☐ (c) IIIIXX ☐ (d) XXVI ☐
11. The correct Roman numeral for 29 is
(a) XXIX ☐ (b) IXXX ☐ (c) XXVIII ☐ (d) IXXL ☐
12. Which of the following Roman numerals is meaningless?
(a) XVIII ☐ (b) IXIV ☐ (c) XXIV ☐ (d) XXIX ☐

QUESTION BAG 2

1. Fill in the missing numerals in the boxes.

Hindu-Arabic Numeral	26		39		13	
Roman Numeral		XVII		XXIX		XIV

2. Compare:

(a) XXXIV $29 + 6$

(b) $16 + 8$ XXVI

(c) XXXIII $50 - 18$

(d) $26 - 19$ IX



3. In each of the following pairs, only one Roman numeral is correct. Circle the correct one.

(a) XXIII, XXIV

(b) XXX, XXL

(c) X, VV

(d) XVIII, XIX

4. Complete the following sequences.

(a) XXIII, , , XXVI, , , , XXX

(b) XV, , , , , XX

(c) XXXIV, , , , XXXVIII, ,

5. Add or subtract and write the answer in Roman numerals.

(a) $V + V =$

(b) $XI - V =$

(c) $XV + IX =$

(d) $XXI - IX =$

(e) $XX - VI =$

(f) $XXXII - VIII =$

6. Write:

(a) the Roman numeral for 50.

(b) a Roman numeral that cannot be repeated.

(c) the Roman numeral for 40.

